

## Practice Set 2: Primary Key (Q1–Q5)

1. Create the hospital database and both tables exactly as shown.

### CODE SCREENSHOT:

```

Query 1
Limit to 10 rows

1 • CREATE DATABASE saval_hospital;
2 • USE saval_hospital;
3
4 • CREATE TABLE doctors (
5     doctor_id INT PRIMARY KEY,
6     dname VARCHAR(50) NOT NULL,
7     department VARCHAR(30)
8 );
9
10 • CREATE TABLE appointments (
11     appt_id INT PRIMARY KEY,
12     doctor_id INT,
13     patient_name VARCHAR(50),
14     fee INT,
15     FOREIGN KEY (doctor_id)
16         REFERENCES doctors(doctor_id)
17         ON DELETE CASCADE
18 );
19 • SELECT * FROM doctors;
20 • SELECT * FROM appointments;
  
```

### DATABASE SCREENSHOT:

| Result Grid         |           |              |      |  |
|---------------------|-----------|--------------|------|--|
| Filter Rows: Search |           |              |      |  |
| appt_id             | doctor_id | patient_name | fee  |  |
| NULL                | NULL      | NULL         | NULL |  |

| Result Grid         |       |             |  |
|---------------------|-------|-------------|--|
| Filter Rows: Search |       |             |  |
| doctor_id           | dname | departme... |  |
| NULL                | NULL  | NULL        |  |

## SQL PRACTICE QNS SOLUTIONS

2. Insert 3 doctors: (1, Dr. Sharma, Heart), (2, Dr. Rai, Bone), (3, Dr. KC, Eye).

### CODE SCREENSHOT:

```
INSERT INTO doctors
VALUES
(1, 'Dr. Sharma', 'Heart'),
(2, 'Dr. Rai', 'Bone'),
(3, 'Dr. KC', 'Eye');
```

### DATABASE SCREENSHOT:

| Result Grid   Filter Rows: |           |            |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|-------------|
|                                                                                                                                                                                              | doctor_id | dname      | departme... |
|                                                                                                                                                                                              | 1         | Dr. Sharma | Heart       |
|                                                                                                                                                                                              | 2         | Dr. Rai    | Bone        |
|                                                                                                                                                                                              | 3         | Dr. KC     | Eye         |
|                                                                                                                                                                                              | NULL      | NULL       | NULL        |

3. Try to insert a doctor with doctor id = 2 again. Write down the error.

### CODE SCREENSHOT:

```
INSERT INTO doctors
VALUES (2, 'Dr. Thapa', 'Skin');
```

| Time     | Action                                              | Response                                                    |
|----------|-----------------------------------------------------|-------------------------------------------------------------|
| 20:41:04 | SELECT * FROM doctors LIMIT 0, 10                   | 3 row(s) returned                                           |
| 20:43:37 | USE saval_hospital                                  | 0 row(s) affected                                           |
| 20:43:56 | INSERT INTO doctors VALUES (2, 'Dr. Thapa', 'Skin') | Error Code: 1062. Duplicate entry '2' for key 'doctors....' |

**ERROR EXPLANATION:** HERE IN THE DOCTORS TABLE PRIMARY KEY IS DOCTOR ID SO IT CANNOT BE REPEATED OR DUPLICATE.

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4. Try to insert a doctor with NULL id, and one with NULL name. Which constraint blocks each?

**CODE SCREENSHOT:**

```
INSERT INTO doctors  
VALUES (NULL, 'Dr. Test', 'Heart');
```

```
INSERT INTO doctors  
VALUES (4, NULL, 'Heart');
```

**EXPLANATION:** *so herefor doctor id primary key cannot be null and in doctor dame (dname) there is not null so this blocks if we try to put null value in each part.*

5. For each real system, decide a good PK and say why name is not enough:  
patients in this hospital, products in Daraz, citizens of Nepal.

**EXPLANATION:**

So in the hospital patient id can be good primary key because there can be same named patient and product id in daraz because it also can have the similar product name and for the citizen of nepal citizenship number can be best because it is unique for every person in nepal.

## Practice Set 2: Foreign Key (Q6–Q12)

6. Insert 4 appointments for existing doctors, two of them for doctor 1. Why is the repeated doctor id allowed?

### CODE SCREENSHOT:

```
INSERT INTO appointments
VALUES
(101, 1, "Ram", 1000),
(102, 1, "Sita", 1200),
(103, 2, "Hari", 800),
(104, 3, "Gita", 900);
```

- *Repeated allowed because the primary key for appointment is appointments id so the doctor id can be repeated.*

### DATABASE SCREENSHOT:

| appt_id | doctor_id | patient_name | fee  |
|---------|-----------|--------------|------|
| 101     | 1         | Ram          | 1000 |
| 102     | 1         | Sita         | 1200 |
| 103     | 2         | Hari         | 800  |
| 104     | 3         | Gita         | 900  |
| NULL    | NULL      | NULL         | NULL |

7. Try to insert an appointment with doctor id = 99. What happens and why is this good?

**CODE SCREENSHOT:**

```
INSERT INTO appointments  
VALUES (105, 99, "saval", 1000);
```

**EXPLANATION:** *It is because there is no id of 99 in doctor table so we can't create it in appointment table as it is child table for doctors table. It makes security and prevent the not necessary appointment from being created mistakenly too.*

8. Which table is parent and which is child? Where does the FK live?

**EXPLANATION:**

*So the parent table is doctor and child is appointment table and the FK live in appointment table doctor id as it is origin is from doctors table which is parent table.*

9. Delete doctor 1. What happened to doctor 1's appointments, and which FK option caused that?

**CODE SCREENSHOT:**

```
DELETE FROM doctors  
WHERE doctor_id = 1;
```

**EXPLANATION:** *BECAUSE OF ON DELETE CASCADE IT IS AUTOMATICALLY DELETED ONCE IT IS DELETED FROM THE PARENT TABLE WHICH IS THE DOCTOR TABLE SO AUTOMATICALLY REMOVED IN APPOINTMENT TABLE.*

10. Recreate the FK with RESTRICT instead. Now try deleting a doctor who has appointments. What happens?

**CODE SCREENSHOT:**

```
ALTER TABLE appointments  
DROP FOREIGN KEY appointments_ibfk_1;
```

```
ALTER TABLE appointments  
ADD CONSTRAINT fk_doctor  
FOREIGN KEY (doctor_id)  
REFERENCES doctors(doctor_id)  
ON DELETE RESTRICT;
```

```
DELETE FROM doctors  
WHERE doctor_id = 2;
```

**DATABASE SCREENSHOT:** SO NOW THE DOCTORS WITH APPOINTMENT IN APPOINTMENT TABLE CANNOT BE DELETED.

11. Real life: Daraz has customers and orders. Draw the parent, the child, and the FK column.

**Code:**

```

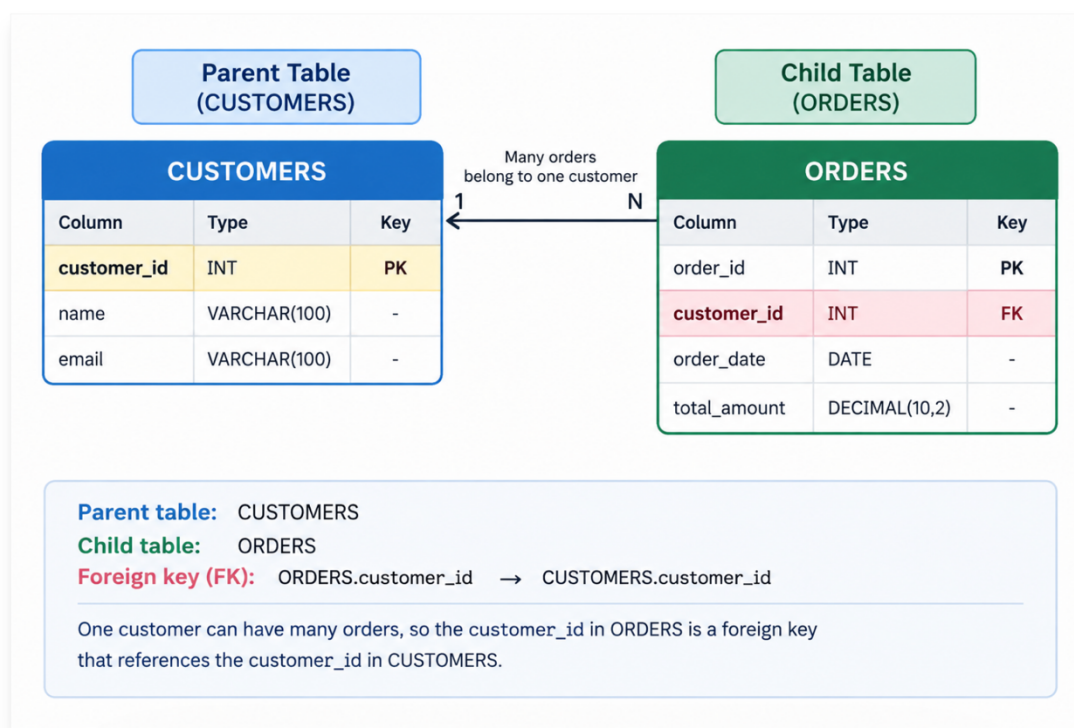
• CREATE TABLE customers (
    customer_id INT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100)
);

• CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    total_amount DECIMAL(10,2),

    FOREIGN KEY (customer_id)
        REFERENCES customers(customer_id)
);

```

**Explanation fig:**



12. Why can an FK repeat and be NULL while a PK cannot? One sentence each.

**FK can repeat:** because many child table can refer to the parent like many appointments belonging to one doctor.

**FK can be NULL:** because when that row currently does not referor link to a parent table.

**PK cannot:** because must uniquely identify every record so it must be unique and cannot be null.